Cape Peninsula University of Technology

Course:	**INFORMATION MANAGEMENT 2**
Course Code:	**INM262S**
Lecturer:	**Leon Small**
Assignment:	**Term 4 Group Project**
Total Marks:	**100**
Students:	**Jody Kearns – 209023651** **Curstin Jade Rose – 220275408** **Simphiwe Nation Kahlana – 220162891** **Thabiso Matsaba – 220296006**
Group:	**Group 4**

Assignment tasks

Curstin Jade Rose

- Completed the introduction of the report
- Worked on the SQL part of the assignment (section 2)

Jody Kearns

- Completed the body of the report
- Worked on the SQL part of the assignment (section 2)
- Formatting of the report

Thabiso Matsaba

- Completed the conclusion of the report
- Worked on the SQL part of the assignment (section 2)

Simphiwe Nation Kahlana

- Completed the body of the report
- Worked on the SQL part of the assignment (section 2)

Cape Peninsula University of Technology
Department of Information Technology
Plagiarism Declaration

Subject: Information Management 2 (INM262S)

Due Date: 22 October 2021

Lecturer: Leon Small

I am submitting the assignment for:

- ☐ an individual project or
- ☒ a group project on behalf of all members of the group. It is hereby confirmed that the submission is authorized by all members of the group, and all members of the group are required to sign this declaration.

I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) if applicable, the submitted soft copy of the assignment is identical to the hard copy to be submitted.

I/We also acknowledge that I am/we are aware of CPUT policy and regulations on plagiarism and honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the CPUT website
https://www.cput.ac.za/storage/study_at_cput/STUDENT_Rules__Regulations_2019.pdf

In the case of a group project, we are aware that each student is responsible and liable to disciplinary actions should there be any plagiarized contents/undeclared multiple submission in the group project, irrespective of whether he/she has signed the declaration and whether he/she has contributed directly or indirectly to the problematic contents.
It is also understood that assignments without a properly signed declaration by the student(s) concerned and in the case of a group project, by all members of the group concerned, will not be graded by the lecturer(s).

Name	Student ID	Signature
Curstin Jade Rose	220275408	
Jody Kearns	209023651	
Simphiwe Nation Kahlana	220162891	
Thabiso Matsaba	220296006	

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Introduction

What is a database

In the information age, data is what drives our society. It can be used to reason, calculate, and discuss. Data can also improve people's lives, help organizations make informed decisions and find solutions to problems. To store data, we utilize databases.

A database is a collection of integrated records, where a record is a representation of a conceptual object. A database is self-explanatory in describing its own structure. This is called metadata, data about data. Databases include relationships with data items, as well as the data items itself (Berg, et al., 2013).

What is a relational database

The relational database model was discovered by E.F. Codd in 1970. Codd's system can be defined using two terminologies (Berg, et al., 2013). A table with rows and columns and a schema that defines the structure.

SQL and the historical evolution of SQL

In 1974, Raymond Boyce and Donald Chamberlin, developers of SQL Server, published an article SEQUEL: A Structured English Query Language, which presented SQL to the world. The publication allowed different vendors to implement SQL into their own database engines (Kozubek-Krycuń, 2020). It then gained popularity when the American National Institute (ANSI) made the first SQL standard in 1986 (The Editors of Encyclopaedia Britannica, 2021).

SQL-86 was the first SQL standard published in 1986 as ANSI standard and was later recognized as International Organization for Standardization (ISO) standard. The ISO standard was IBM's SQL standard implementation, this version was also known as SQL 1.

Following SQL-86, the next SQL standard was SQL-89 which was published in 1989. This release was a minor revision on its predecessor. The initial size of the standard document did not change.

In 1992, the next revision was SQL-92. This was a major version and is sometimes referred to as SQL 2. The size of the standard document increased as many existing features were improved upon. SQL-92 is also the foundation of query languages used in relational databases. It is what most people refer to when teaching SQL today.

SQL:1999, also known as SQL 3, was the fourth release of the SQL standard. Along with this standard, the standard name used a colon to be consistent with the other ISO standard names. This standard was published in multiple instalments between 1999 and 2002.

Today, the SQL standard has been regularly updated. Each revision of the SQL standard attempts to align itself with modern technology needs. Such as support for window functions, XML-related functions, and JSON-support. SQL:1999 and SQL:2003 established the foundation for modern SQL.

Importance of SQL in modern computing

SQL in the modern era

Structured query language, also known as SQL, is currently one of the most essential, valuable and mainstream languages used to access and manage databases in the modern era. Regardless of the size of the company, whether big or small, SQL is used by companies to interact with their databases and access data and information that their databases hold. (Babu, 2018) The ability to effectively mine these big quantities of data and present them in a format that is both easy to understand and informative is crucial to the success of these businesses. SQL is useful when it comes to doing this as it is able to easily take large data sets and process the data in a more robust and faster manner compared to other data applications such as Excel while also ensuring data integrity. (Davies, 2021)

The biggest uses for SQL in recent years are using it along with relational databases in emerging IT enterprise environments such as cloud-native systems and virtual systems. (Stoltzfus, 2020) As these systems carry huge amounts of data that increase from day to day, SQL displays why it is beneficial by providing the flexibility and scalability that is needed. (Indeed, 2021)

The demand for SQL

In 2021, Indeed.com placed SQL on the list "11 of the Most In-Demand Coding Languages", this speaks to the importance of SQL as businesses are looking in abundance for people that hold this specific skillset. (Indeed, 2021)

Data Science and its need for SQL

As the current world is in the data driven direction, with Artificial intelligence and Machine learning on the rise, data science is hugely dependant on the gathering and processing of data. SQL is first on the list in terms of managing the data involved with data science as SQL allows the necessary skills to the job duties you would need to access, manipulate search and control datasets. (Mutai, 2021)

Below is a table taken from "towards data science" which illustrates the 10 most in-demand data science skills in 2021:

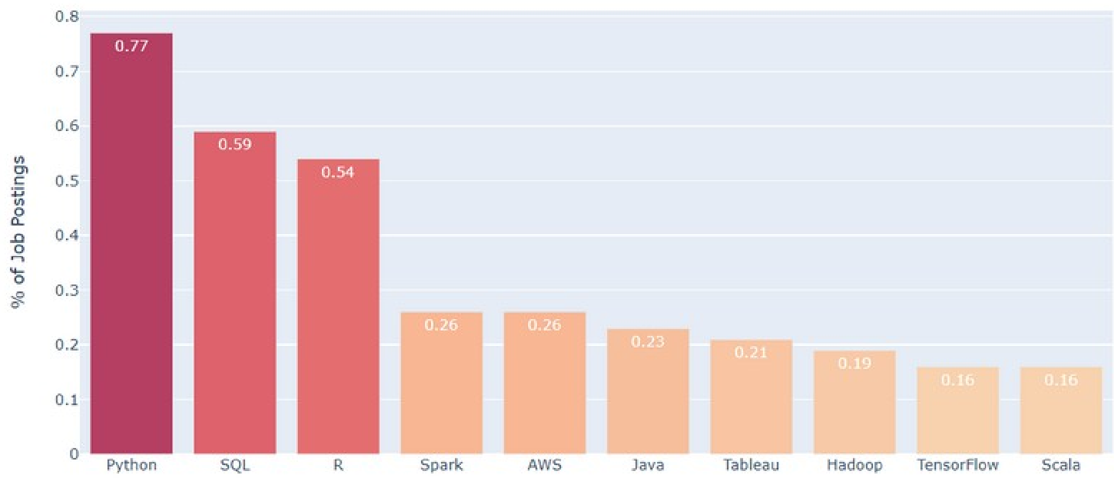

Figure 1: 10 Most In-Demand Data Science Skills in 2021

As seen in the above graph, SQL is the second most in demand skillset in the data science field, which is one of the most emerging fields, coming only second to the Python programming language. (Shin, 2021)

Even though SQL is second to Python, it is important to acknowledge how versatile SQL is as it gets utilised in conjunction with various programming languages such as Python, Java, R and C++ to do database performance analysis, develop effective cross-platform software, support data storage and handles the retrieval of data and processing it to the front end application in server client requests.

SQL in modern web development

Most modern websites are interactive and make use of SQL to manage the data that the user interacts with on these websites. As a website has its own database, SQL is used to retrieve this data that is stored in the database and allows users of the website to interact with the data as needed. (Chakraborty, 2021)

A common use case that displays the importance of SQL is its use in the ecommerce sector. Amazon is the biggest company in the world when it comes to ecommerce and they make use of databases to store customer information, seller information and information about their products on offer. While from a frontend perspective it might seem to the user that this information is already stored on the website itself, in the backend, SQL is used to query the databases where this information is stored in and then retrieve the requested information via SQL which will return it to the front end and then display it for the user to see and interact with. With a company as big as Amazon making use of SQL to provide a service to their clients, it points to the importance of SQL in the modern era of computing.

SQL for the future

Based on the above it is evident that SQL is a cornerstone of the programming world, whether it is a small application that uses data stored in a database or a huge $1.7 trillion dollar company that needs to make use of SQL to handle their data. SQL does and will play an important role for the future of data science and the progression of data science too. While many people might say that SQL is becoming redundant, the evidence is clear that it is one of the most integral computing languages in the current era of data.

Reflections and value of LinkedIn learning

Curstin Jade Rose

What I have learnt about SQL is that it is a versatile programming language that is able to integrate seamlessly into any application or program. It gives users the ability to access or query data fast without having extensive knowledge on SQL. SQL data definition allows for structured data and being able to manipulate data in a way we see fit is what makes it an excellent choice to implement in your systems. Learning SQL through LinkedIn Learning has it's benefit as it introduces the topic in a well-presented video format. Professionals in the industry teaching the course are excellent as they give insight into industry standards and good practice. However, it is not something I will refer to if I need to recall a specific part of

SQL. Reading the specific part online would be a lot faster than watching the video from the start.

Thabiso Matsaba

I have learnt a lot about SQL and its application in the modern world. Businesses, IT administrators, and Software developers need to have a sound knowledge of SQL, because they will always have to work with databases. SQL has been the language of choice for many of these professions and still is in use and in high demand even today.
LinkedIn learning was a very good and informative platform to learn SQL. The way in which the videos have been created makes it easy to follow and the quizzes are a good way to grasp the knowledge presented. The LinkedIn Learning courses not only helped me with understanding SQL only, but other subjects as well. The certificates I have got through LinkedIn Learning will also help me in the future when applying for employment.

Jody Kearns

What I've learnt this year is that SQL is an extremely vital part of the programming world as it is used in all applications and systems that has a database and requires data to be handled. I've also learnt that SQL is extremely flexible as it integrates with many other programming languages such as Python and Java, which are languages that I am interested in pursuing.
LinkedIn learning provided a strong platform for me to utilise to learn concepts and syntax of SQL on my own. What I liked about LinkedIn learning too, is that it added what you have learnt to your LinkedIn profile which will help in the future as recruiters will be able to see which courses I took and see my skillset at a glance. LinkedIn learning definitely helped me understand SQL easier as the course I completed had exercises to do which is a learning style that I thrive at.

Simphiwe Nation Kahlana

The course lectures directed us to 5 different online courses on LinkedIn. These courses have taught me how to write data retrieval queries and evaluate the results, how to write SQL statements that edit existing data and how to write SQL statements that create database objects. These self-paced courses were very helpful as I now also understand the structure and design of relational databases and the importance and major issues of database security and the maintenance of data integrity. Besides these courses being self-paced, what I also liked is that I could repeat a section as many times as I liked until I understood it. The LinkedIn courses also offered exams/tests after every learning objective which also helped in testing the knowledge I was learning throughout the courses.

Conclusion

SQL today remains the backbone of every relational database and many non-relational databases. Even though there is a rise in Big Data and Cloud computing, SQL remains the most in demand programming language for relational databases. This is the reason why it is important to learn how to code using SQL, as it will remain in place for years to come, as the standard data access language used across many fields and practises.

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Meeting Minutes

Date: 02 October 2021 & 19 October 2021
Time: 11:00 AM
Facilitator: Jody Kearns
Online Meeting Platforms
Blackboard Collaborate
Attendee List
- Jody Kearns
- Curstin Rose
- Simphiwe Kahlana
- Thabiso Matsaba

Absent List

Meeting 1:
Activities
- Discuss on how to complete the project
- Section 1 (The project report)
- Introduction of the report
- Body of the report
- Reflections
- References (min 6 sources)
- Division of Duties

Discussion
Firstly, each member introduced themselves. After that, the facilitator started to go through the brief and the group discussed. The group focused on Section 1 (The project report) and listed all the points to complete.
- Introduction
  - Historical evolution of modern-day SQL
- Body
  - Research/Presentation on importance of SQL
  - Graphs/Charts/Tables on SQL
  - List companies that use SQL
- Reflection
  - What you've learnt about SQL
  - Value of LinkedIn learning in your journey in the course.
  - Each group member must write up their own experience.
- Conclusion
  - We decided that each member must research for information on the topic and get reputable sources to reference. The group decided to use a Google Document for the report, this way everyone can have access to the same document.

Action Items
- Create google doc -> Curstin Rose
- Gather information - > Group
- Section 1 (The project report) -> Group

Meeting 2:
- Completed SQL (Part 2).
- Finalised Report.